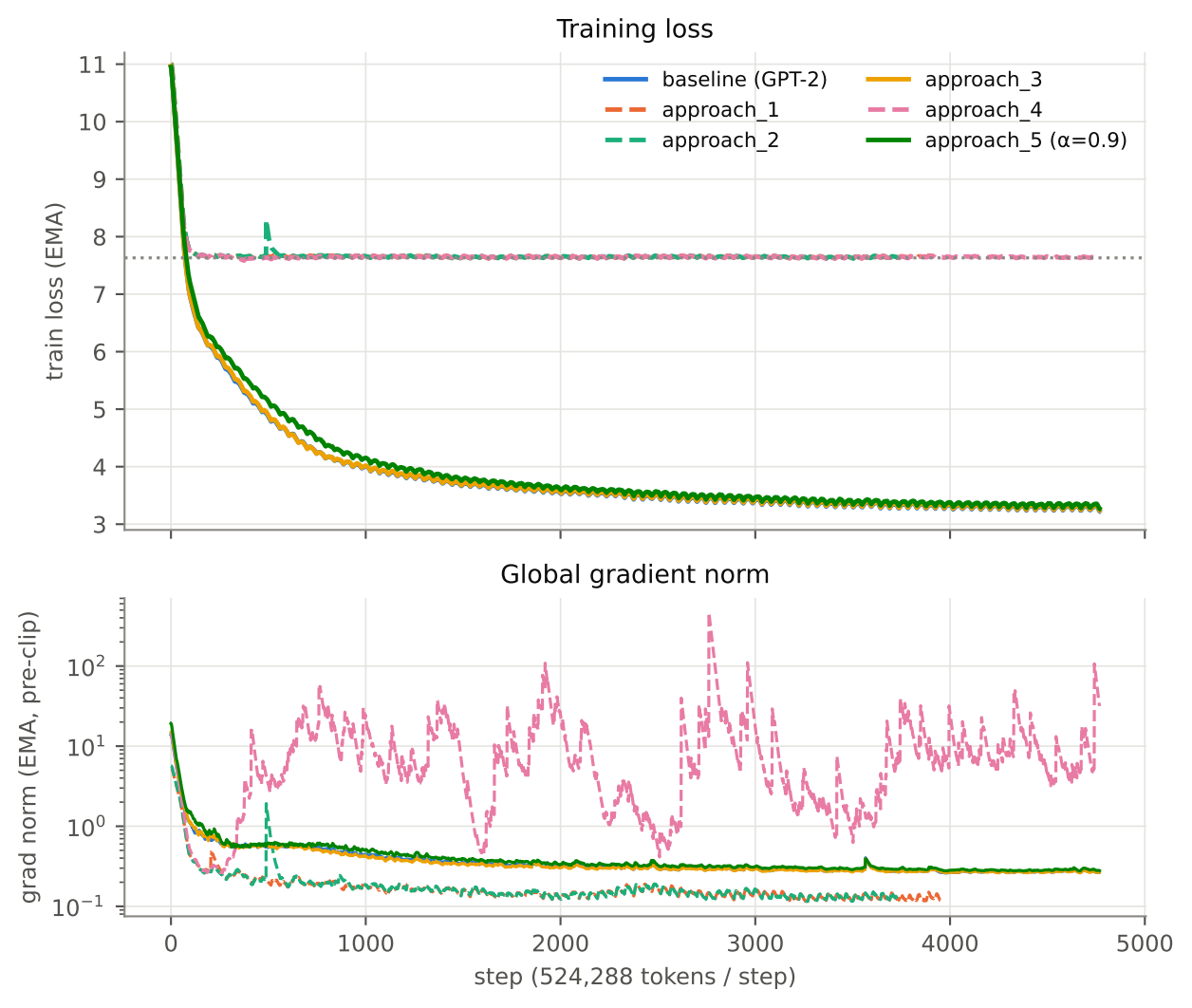

Training loss
baseline (GPT-2)
approach_1
approach_2
approach_3
approach_4
approach_5 (α=0.9)
train loss (EMA)
11
10
9
8
7
6
5
4
3
Global gradient norm
grad norm (EMA, pre-clip)
10^2
10^1
10^0
10^{-1}
0
1000
2000
3000
4000
5000
step (524,288 tokens / step)